

EBIO 5460

Process Modeling with Data for Ecology & Evolution

Dr Brett Melbourne
Professor, EBIO

brett.melbourne@colorado.edu

Office hours: Any time by appointment

Office: Ramaley N336 and Zoom

Pronouns: he, him, his

Git & GitHub

- Class Github organization
- Bookmark this:
- <https://github.com/EBIO5460Fall2026>
- Organization, syllabus, timetable
- Slides, code, homework
- You'll also submit your work here
- Main resource links: [README.md](#)

Slides for today

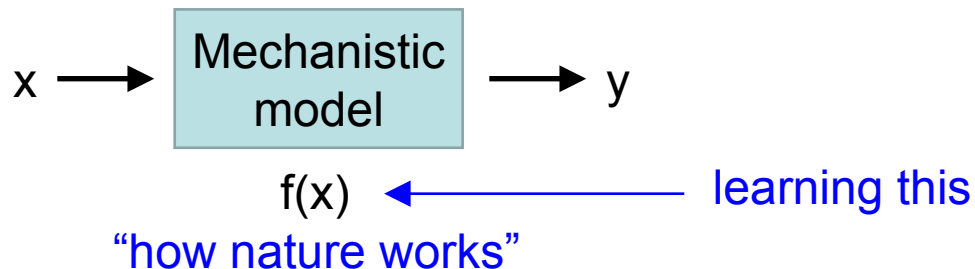
- github.com/EBIO5460Fall2026
- Go to repositories
- Open class-materials
- 01_1_slides_thu_intro

Today

- Intro to process models (2 mins)
- Introductions (20 mins)
- Syllabus & how we'll do the class (20 mins)
- Where does process modeling fit in to data science & algorithms?

What is a process model?

- Working definition
- Includes mechanistic biological processes to predict outputs y from inputs x
- Emphasis: describes biology with fidelity
 - e.g. processes: birth, death, movement, search, decision, consumption, growth, infection, mutation, recombination



Examples in ecology

- Population dynamics in fragmented landscapes
 - predicting the spatial patterns of beetle species from **births**, **deaths**, and **movement**
- What are your examples?

Introductions

- Name (and pronouns)
- Masters or PhD (what year)?
- Advisor
- Department
- What fascinates you (your research)?
- Hopes for the course

Syllabus

- We worked through it here:
- [00_syllabus.md](#)

Learning goals

- Understand the **fundamental concepts** and **algorithms** that underpin most process models
- Become **confident** to derive process-based algorithms in **your research**
- Gain a **broad overview** of how process models are currently used in EEB

Learning format

- Coding demonstration in live lectures, sometimes short videos
- Small coding assignments to practice fundamental concepts and skills
- Critically read current papers to practice recognition/understanding and inspiration
- Make a process model for a real-life, messy research system (yours) to integrate and apply concepts and skills

Collaborative learning

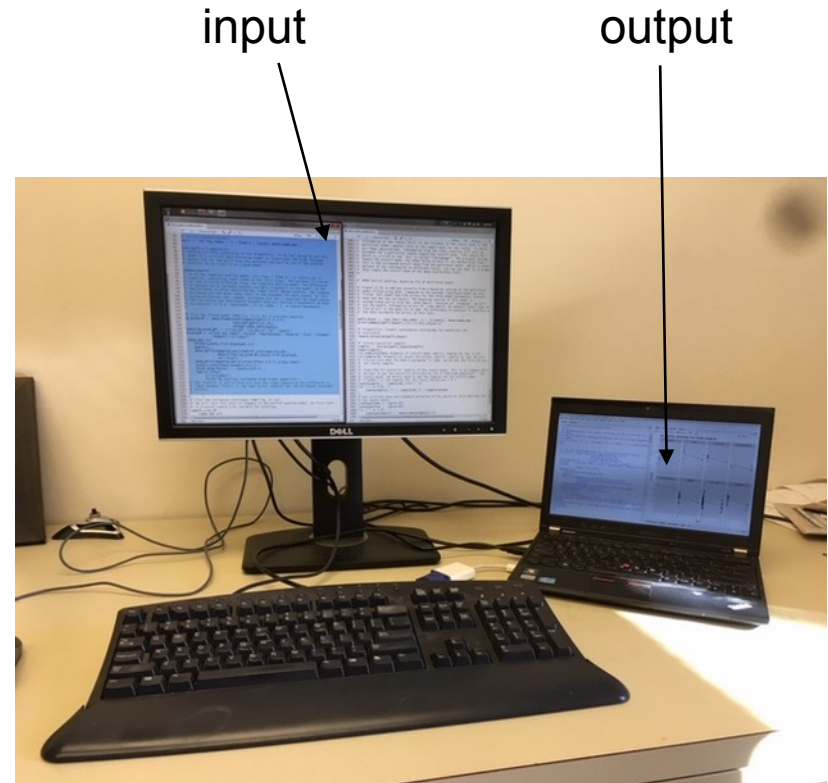
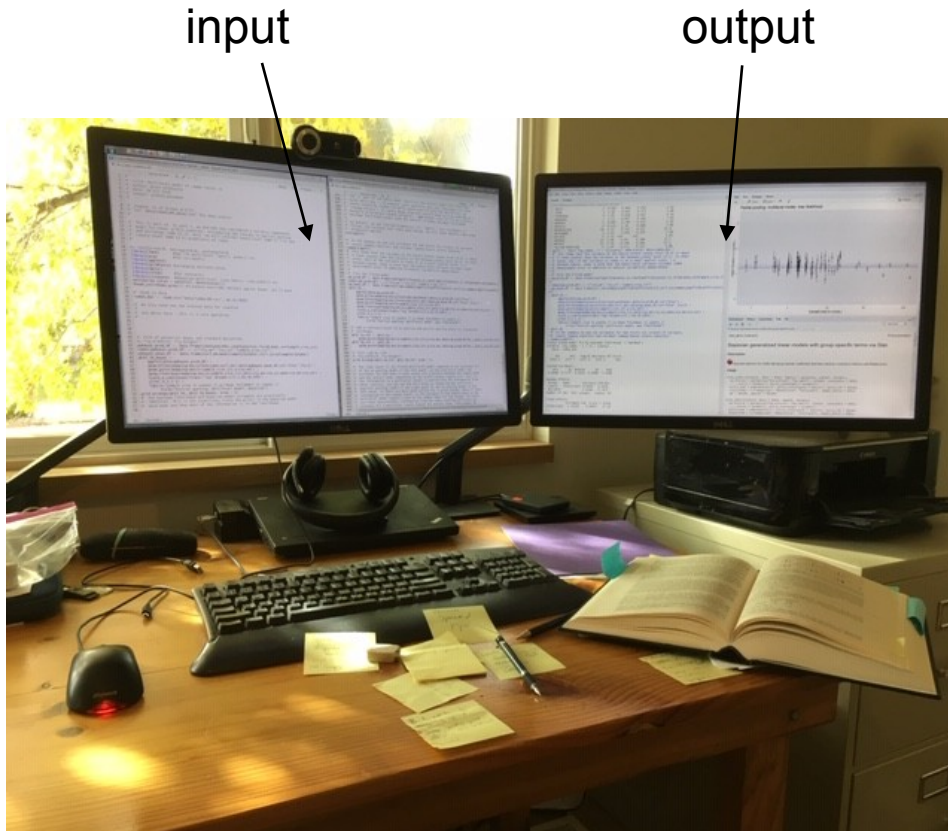
- Collaborative learning is not only allowed but **encouraged** in this class!
- Work in small groups or share in small groups
- **Piazza**: do we want to do this?
 - Collaboratively discuss lectures and assignments. [Link @ README.md](#). FERPA compliant.

Computing

- Install/Update R ... and/or
- Install/Update Python (suggest: via conda)
- Install/Update IDE(s)
 - e.g. Positron, Rstudio, VSCode

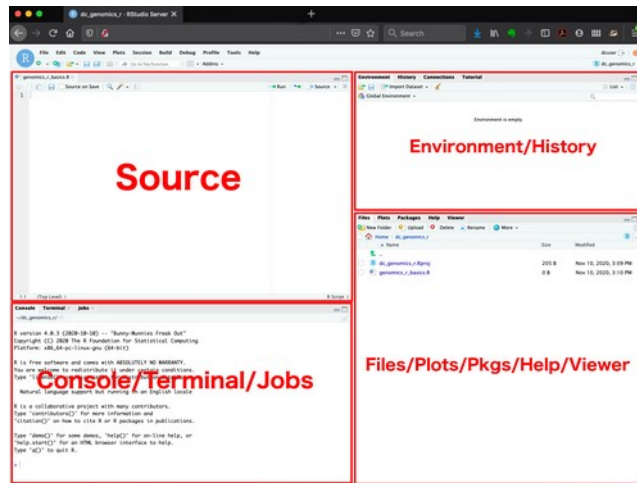
Productive workspaces

side by side layout

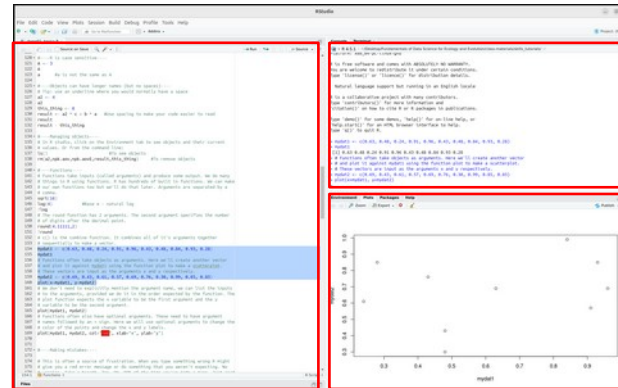


Multiple monitors are essential!

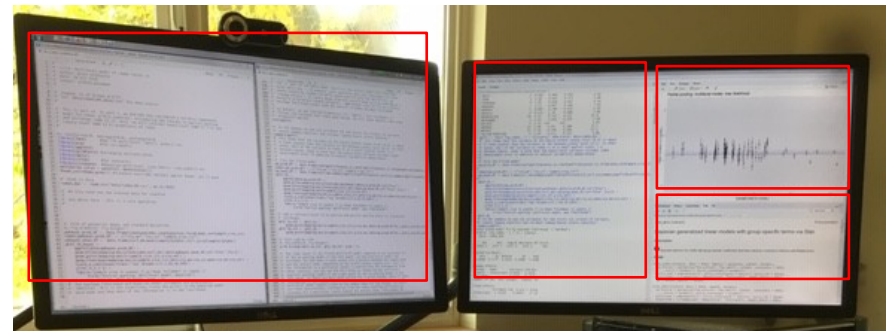
Productive workspaces stacked



side by side



multimonitor side by side



least productive



most productive

Text

- None
- I wish there was
- I'll try to write one
- I'll draw from lots of places
- I'll add a list of some of my favorites

Grading

- For completion
- Assignments 35%
- Discussions 20%
- Lead discussion 10%
- Individual project 35%

Week 1-10 assignments

- Will be posted to GitHub
- [01_3_homework_to_get_started.md](#)
 - Learn Git (if needed)
 - Review algorithms (if needed)
 - Set up GitHub (required)
- Week 1 HW is not part of grade but needs to be done by Tuesday Week 2
- 5 assignments: $5 \times 7\% = 35\%$

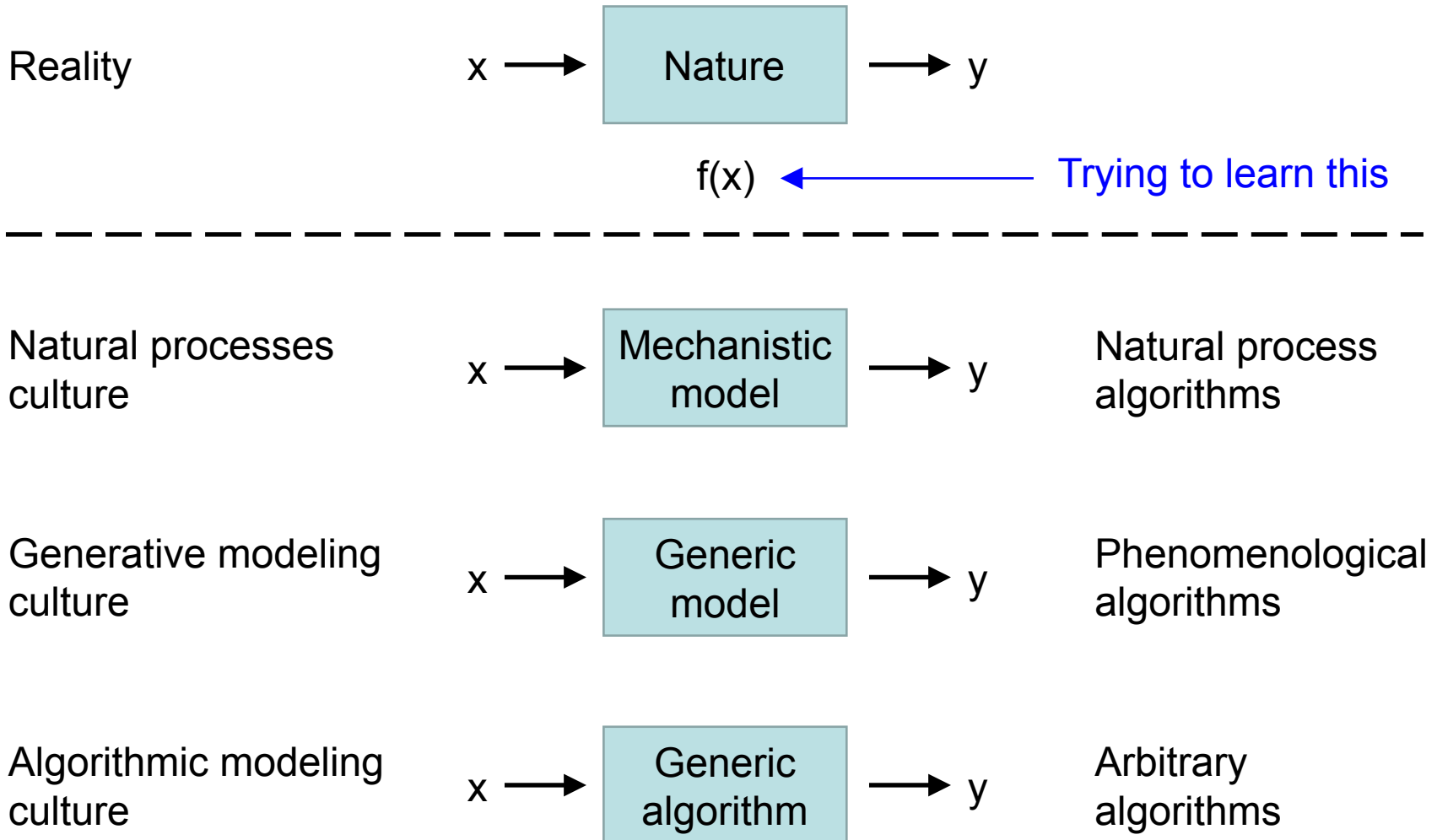
Week 11-15 literature & project

- Each person will lead a discussion 10%
 - paper of your choice
 - ca 2% per discussion (reflection, participation)
- Individual project
 - process model(s), optionally with data
 - system and data set of your choice
 - presentation in finals week 15%
 - code or paper submission 20%

Data Science

- Workflows and **algorithms** to learn from data
- Part 1 (Fall semester 2025): Fundamental algorithms and concepts
 - <https://github.com/EBIO5460Fall2025/class-materials> (also algorithms compilation)
- Part 2 (Spring semester 2026): Machine learning
- Part 3 (this class): Process models

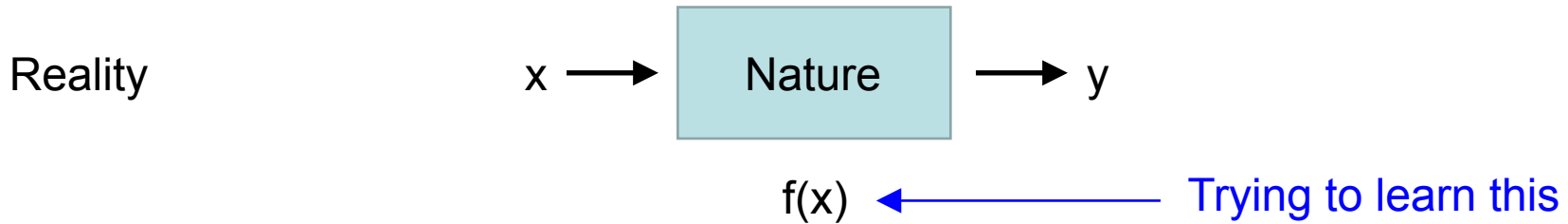
Data science cultures



f can mean different things in different cultures

Breiman (2001)
Denoho (2017)

Data science cultures



Science (e.g. ecological model)

Natural processes
culture

$x \rightarrow$

Mechanistic
model

$\rightarrow y$

Natural process
algorithms

Statistics (e.g. GLMM)

Generative modeling
culture

$x \rightarrow$

Generic
model

$\rightarrow y$

Phenomenological
algorithms

Machine learning/AI (e.g. neural network)

Algorithmic modeling
culture

$x \rightarrow$

Generic
algorithm

$\rightarrow y$

Arbitrary
algorithms

f can mean different things in different cultures

Breiman (2001)
Denoho (2017)

Algorithm

- **Procedure** for solving a problem or describing a system in terms of **actions** to execute and **order** to execute them
- **Code, computation**
- Algorithms are **fundamental**
 - most math in statistics is a solution or approx to a **data-generating** algorithm
 - **biological processes** are naturally **algorithmic**

Algorithms in data science

- Model algorithm
- Training algorithm
- Inference (reliability) algorithm

Algorithms in data science

- Model algorithm
 - The function $f(x)$!
 - Often equations, sometimes rules
 - Usually has parameters
 - e.g. $y = a + b x$
- Training algorithm
- Inference (reliability) algorithm

Algorithms in data science

- Model algorithm
- Training algorithm
 - Algorithm to train a model algorithm on data
 - syn. model fitting, calibration, parameter estimation
 - e.g. Nelder-Mead simplex optimization, gradient descent, Rosenbluth (MCMC)
- Inference (reliability) algorithm

Algorithms in data science

- Model algorithm
- Training algorithm
- Inference (reliability) algorithm
 - first, what kind of inference?

Statistical inference

- Judge the **accuracy** of an estimation or prediction algorithm
 - Efron & Hastie 2016
- **Reliability**
- **Uncertainty**

ISO definition of accuracy: the closeness of a measurement to the true value
Two components: bias, variance

Different inference problems

Estimation

Infer a property of a population (e.g. growth rate) from a sample

Model comparison

Infer the data generating process from among a set of candidate data-generating processes (hypotheses/models)

Hypothesis test of association

Infer that y is associated with x

Causation

Infer that x causes y

Infer the size of an effect due to an experimental intervention (estimation)

Infer that an experimental intervention had an effect (H-test)

Prediction

Predict the value of a new observation or population state (extrapolation or interpolation)

Predict the population state in the future (forecast/extrapolation)

Algorithms in data science

- Model algorithm
 - Training algorithm
 - Inference (reliability) algorithm
 - **looking back**: consider all the ways data could have happened (**mechanistic**, generative)
 - **looking forward**: predict new data and test against them (**mechanistic**, generative, algorithmic)
- Process models

Algorithms review

`algorithms4ds_review.md`

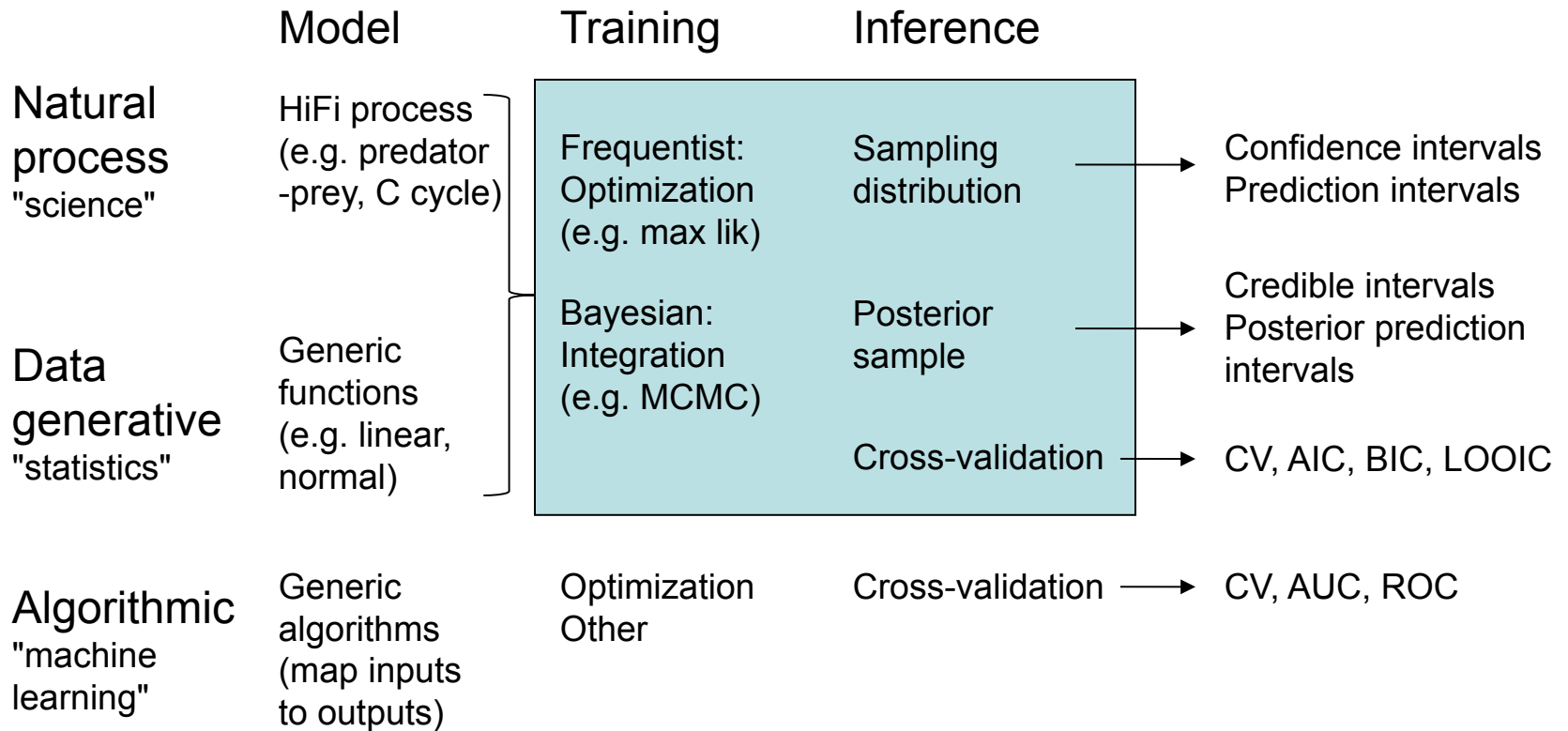
See also algorithms compilation:

https://github.com/EBIO5460Fall2025/class-materials/blob/main/00_fundamental_algorithms.md

Modeling with data

Algorithm classes

Modeling culture



Process models

- From Fundamentals of Data Science:
- Organism movement by random walk
- Logistic growth
 - deterministic, fitted with SSQ
- Squirrel finding nuts